

NEURAL NETWORKS

Foundations: Neural Networks to Representation Learning

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UNIVERSIDAD DISTRITAL
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- 1 Neural Network Fundamentals
- 2 Optimization & Regularization



Outline

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1 Neural Network Fundamentals

2 Optimization & Regularization

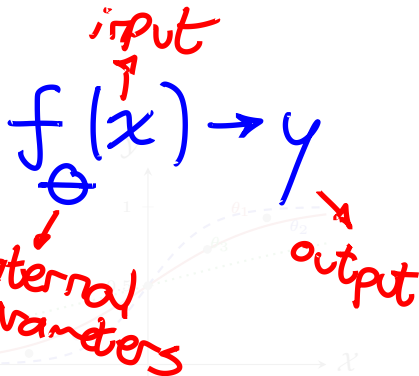


What is a Neural Network?

Definition

A neural network is a parametric function $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ composed of simple, interconnected computational units (neurons).

- **Neuron:** weighted sum + nonlinear activation
- **Layer:** neurons operating in parallel
matrix: x
↓
linear equations
- **Depth:** key enabler of hierarchical learning
- **Training:** adjusting θ so that $f_{\theta}(x) \approx y$



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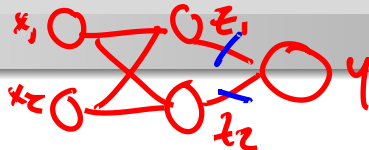
$$- w_3 x_1 + w_4 x_2 = z_1$$

$$w_3 x_1 + w_4 x_2 = z_2$$

$$y = w_5 z_1 + w_6 z_2$$

$$= w_5 w_3 x_1 + w_5 w_4 x_2 + w_6 w_3 x_1 + w_6 w_4 x_2$$

$$= (w_5 w_3 + w_6 w_3) x_1 + (w_5 w_4 + w_6 w_4) x_2$$



$$z_0 = \underbrace{w_1 x_1 + w_2 x_2 + \dots}_{\text{weights}}$$

↓

more inference

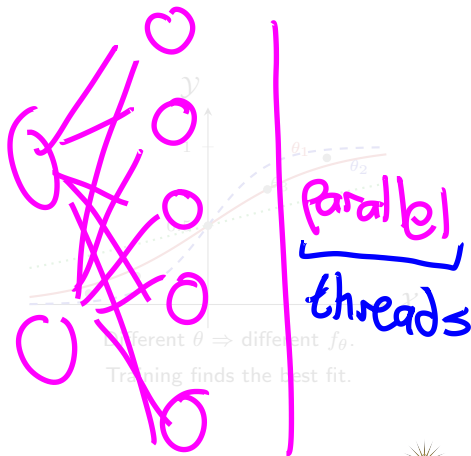


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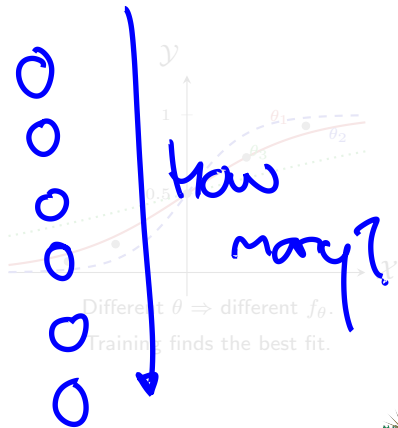
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How many layers?



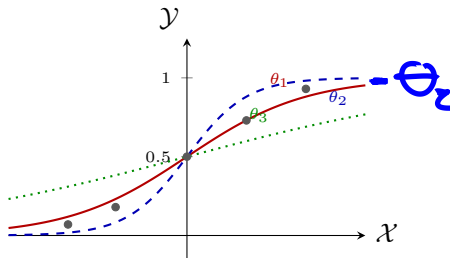
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Different $\theta \Rightarrow$ different f_{θ} .

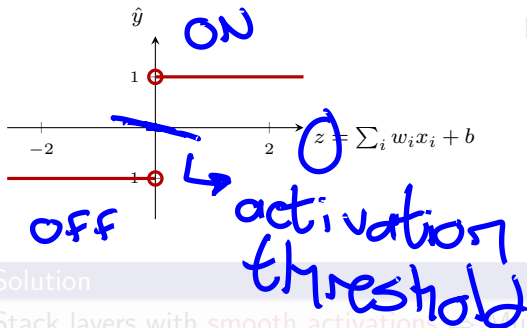
Training finds the best fit.



The Perceptron

Rosenblatt (1958)

$$\hat{y} = \text{sign}(\sum_i w_i x_i + b)$$



Strengths

- Binary classification
- Convergence guarantee (linearly separable data)

Limitations

- Cannot solve XOR
- Only linearly separable
- Hard threshold — not differentiable

Solution

Stack layers with smooth activations → Multi-Layer Perceptron

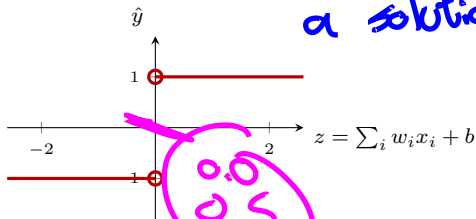


The Perceptron

1957 - AI. *Handwritten: Rosenblatt (1958)*

$$\hat{y} = \text{sign}(\sum_i w_i x_i + b)$$

Handwritten: there is a solution



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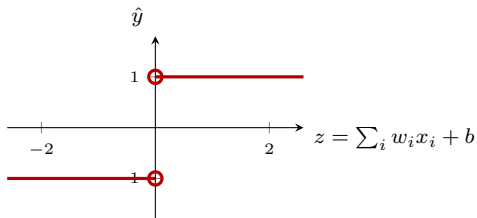
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Multi-Layer Perceptron (MLP) — Architecture

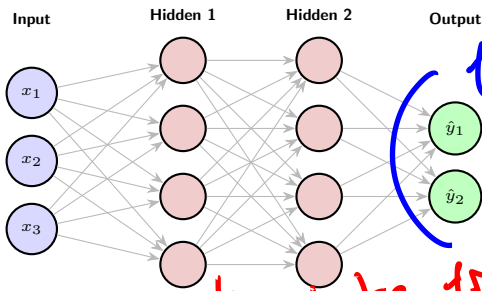
Architecture

Input layer → Hidden layer(s) → Output layer

For each layer l :

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \quad \mathbf{a}^{(l)} = \sigma(\mathbf{z}^{(l)})$$

DAG
↓
Directed
Acyclic
Graph



threshold
"new idea"
→ knowledge transfer



Multi-Layer Perceptron (MLP) — Architecture

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Input layer → Hidden layer(s) → Output layer

- **Universal Approximation** (*Hornik, 1991*): one hidden layer with enough neurons approximates **any** continuous function

• Depth more parameter-efficient than extreme width



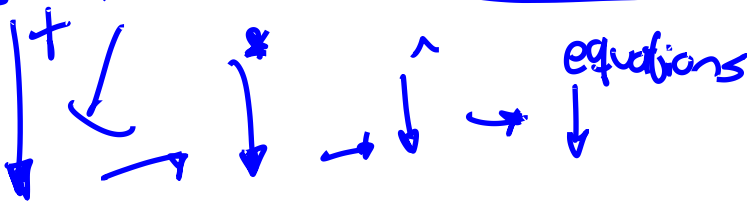
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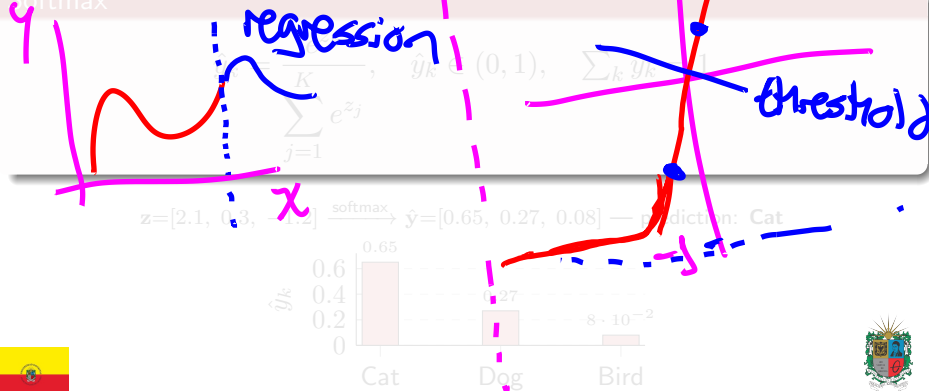


Multi-Layer Perceptron (MLP) — Output & Softmax

Output layer choices

- **Linear**: regression
- **Sigmoid**: binary classification
- **Softmax**: multi-class $\rightarrow$ probability distribution over K classes

Softmax



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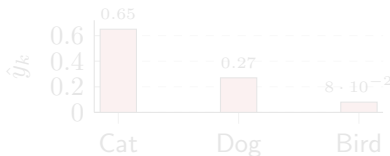
fuzzy
log.c

Softmax

$$\hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad \hat{y}_k \in (0, 1), \quad \sum_k \hat{y}_k = 1$$

convex
combination

$$\mathbf{z} = [2.1, 0.3, -1.2] \xrightarrow{\text{softmax}} \hat{\mathbf{y}} = [0.65, 0.27, 8 \cdot 10^{-2}]$$



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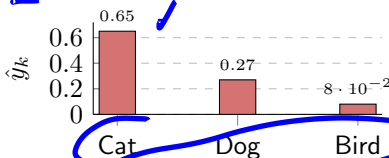
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$z = [2.1, 0.3, -1.2] \xrightarrow{\text{softmax}} \hat{y} = [0.65, 0.27, 0.08] \text{ — prediction: Cat}$



K categories



Activation Functions

Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \in (0, 1)$$

Tanh:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \in (-1, 1)$$

ReLU: $\max(0, x)$ — **default today**

GELU: $x \cdot \Phi(x)$ — used in Transformers

Why ReLU dominates

- No saturation for positive inputs
- Sparse activation (many zeros)
- Computationally trivial
- Mitigates **vanishing gradient**



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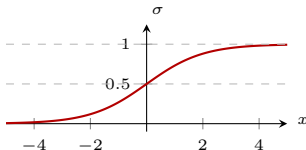
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Activation Functions — *Visual Overview*

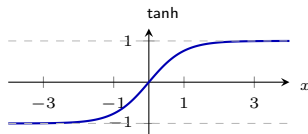
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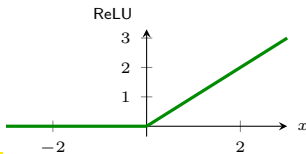
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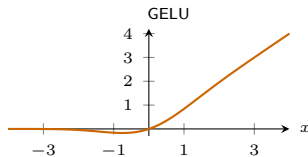
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$$\max(0, x)$$



GELU

$$x \cdot \Phi(x)$$



Loss Functions

The **Objective** of Training

Minimise $\mathcal{L}(\theta)$ — a measure of disagreement between **predictions** and **targets** over the training set.



Loss Functions

Mean Squared Error (*regression*)

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_i - y_i\|^2$$

Categorical Cross-Entropy (*classification*)

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}$$

- Cross-entropy + softmax is the standard for multiclass classification
- The choice of loss determines the gradient signal the network receives



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Backpropagation

Core Idea

Apply the **chain rule** repeatedly to compute $\partial \mathcal{L} / \partial \mathbf{W}^{(l)}$ for every layer, from output back to input.

Two-pass algorithm:

- Forward pass: compute $\mathbf{z}^{(l)}, \mathbf{a}^{(l)}$ for all l ; store activations
- Backward pass: compute error signal $\delta^{(l)}$, then $\nabla_{\mathbf{W}^{(l)}} \mathcal{L} = \delta^{(l)} (\mathbf{a}^{(l-1)})^T$
- Time complexity: $O(n^3)$ — same order as a forward pass
- Made efficient by automatic differentiation (PyTorch, TensorFlow)



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The Training Loop

- 1 **Initialise** weights θ_0 (random)
- 2 Split data: **train** / **validation** / **test**
- 3 For each **epoch**:
 - Sample a mini-batch $\{(x_i, y_i)\}_{i=1}^B$
 - Forward pass $\rightarrow$ compute $\mathcal{L}$
 - Backward pass $\rightarrow$ compute $\nabla \mathcal{L}$
 - Update: $\theta \leftarrow \theta - \alpha \nabla \mathcal{L}$
- 4 Monitor **validation loss**; stop when it increases (early stopping)

Golden Rule

Never tune hyperparameters using the **test set**. The test set measures final generalisation only.



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Practical Example: MLP on Two-Moons

- **Problem:** binary classification on non-linear boundary
- **Architecture:** $2 \rightarrow 32 \rightarrow 32 \rightarrow 1$, ReLU, sigmoid output
- **Experiment:** vary number of hidden layers and width
- **Observation:** deep narrow network generalises better than wide shallow one
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Gradient Descent

Update Rule

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta)$$

Full-batch GD

- Exact gradient
- Expensive per step
- Deterministic

Mini-batch SGD

- Noisy gradient estimate
- Cheap per step
- Noise helps escape local minima
- Standard in practice

- Batch size B : typical values 32–512; affects convergence and memory
- Learning rate α is the most critical hyperparameter



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$$\mathbf{v} \leftarrow \beta \mathbf{v} - \alpha \nabla \mathcal{L}, \quad \theta \leftarrow \theta + \mathbf{v}$$


Momentum and Nesterov

SGD with Momentum

$$\mathbf{v} \leftarrow \beta \mathbf{v} - \alpha \nabla \mathcal{L}, \quad \theta \leftarrow \theta + \mathbf{v}$$

- Typical $\beta = 0.9$; accumulates gradient **history**
- Dampens oscillations; accelerates in consistent directions
- **Nesterov**: compute gradient at the **look-ahead** position $\theta + \beta \mathbf{v}$ instead of current θ — slightly better convergence

Intuition

Momentum is like a ball rolling down a hill that builds **inertia**. Nesterov is a ball that **looks ahead** before deciding where to roll.



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Adam Optimizer

Adaptive Moment Estimation (Kingma & Ba, 2015)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$\hat{m}_t = m_t / (1 - \beta_1^t)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{v}_t = v_t / (1 - \beta_2^t)$$

$$\theta \leftarrow \theta - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$$

• Defaults: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, $\alpha = 10^{-3}$

• Per-parameter adaptive learning rates

AdamW: Adaptive Moment Estimation with Weight Decay

PyTorch

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Adam Optimizer

Adaptive Moment Estimation (Kingma & Ba, 2015)

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 m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t & \hat{m}_t &= m_t / (1 - \beta_1^t) \\
 v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 & \hat{v}_t &= v_t / (1 - \beta_2^t) \\
 \theta &\leftarrow \theta - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)
 \end{aligned}$$

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- Per-parameter **adaptive learning rates**
- **AdamW**: Adam + decoupled weight decay — preferred in current practice
- Adam dominates in deep learning; SGD+Momentum preferred in CV research



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Adam Optimizer

Adaptive Moment Estimation (Kingma & Ba, 2015)

$$\begin{aligned}
 m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t & \hat{m}_t &= m_t / (1 - \beta_1^t) \\
 v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 & \hat{v}_t &= v_t / (1 - \beta_2^t) \\
 \theta &\leftarrow \theta - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)
 \end{aligned}$$

- Defaults: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, $\alpha = 10^{-3}$
- Per-parameter **adaptive learning rates**
- **AdamW**: Adam + decoupled weight decay — preferred in current practice
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Learning Rate Scheduling

- **Step decay:** multiply LR by factor γ every k epochs
- **Cosine annealing:** $\alpha_t = \alpha_{\min} + \frac{1}{2}(\alpha_{\max} - \alpha_{\min})(1 + \cos(\pi t/T))$
- **Linear warmup:** ramp from near-zero for first w steps, then decay
- **ReduceLROnPlateau:** halve LR when validation loss stops improving

Warmup and Decay

Modern practice: **linear warmup** + **cosine decay**. Used in ResNets, BERT, and GPT. Especially important with large batch sizes.



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Underfitting (high bias)

- Model too simple
- High training error
- Cannot capture data structure
- Fix: increase **capacity**

Overfitting (high variance)

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- Memorises noise
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Practical Guideline

Start with a model that **slightly overfits**, then regularise until train and val errors converge. Never start with a model that underfits.



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L1 and L2 Regularization

Augmented Objective

$$\mathcal{L}_{\text{reg}}(\theta) = \mathcal{L}(\theta) + \Omega(\theta)$$

L2 Regularization (Weight Decay / Ridge)

$$\Omega(\theta) = \frac{\lambda}{2} \|\mathbf{W}\|_F^2 \quad \text{drives weights small but rarely zero}$$

L1 Regularization (Lasso)

$$\Omega(\theta) = \lambda \|\mathbf{W}\|_1 \quad \text{promotes sparse weights (many exact zeros)}$$

• In deep learning, L2 (weight decay) is the standard choice
 • Both regularization methods — used in linear models and neural networks



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Dropout

Definition (Srivastava et al., 2014)

During training, independently zero out each activation with probability p . At inference, multiply activations by $(1 - p)$ (*inverted dropout* scales during training).

- Forces redundant representations — no single neuron can be relied upon
- Equivalent to averaging over 2^n thinned networks
- Typical $p = 0.5$ (hidden layers), $p = 0.1$ after convolutional layers
- *Dropout* is a special case of *inverted dropout* with the same p for all layers



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Normalise activations within a mini-batch, then **scale and shift**:

$$\hat{x}_i = \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta$$

- γ, β are learnable parameters
- Stabilises training: allows larger learning rates
- Reduces sensitivity to initialisation
- Allows for more effective use of variance accumulated during training
- Works better with smaller batch sizes



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- **Problem:** overfit a 5-layer MLP on a small dataset
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Outline

- 1 Neural Network Fundamentals
- 2 Optimization & Regularization



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning>

